

Lung Nodule Segmentation — DynUNet / 3D U-Net — sliding-window (128^3), CE weight 100

1. Task

Voxel-level segmentation of pulmonary nodules in 3D chest CT volumes, lung-bbox-cropped. Same task as the earlier resize-recipe `dynunet` model — the only difference is inference recipe: **sliding-window over 128^3 patches at native resolution** rather than a single forward on a 256^3 resample.

- **Input** (1, H, W, D) CT crop, intensity-normalised to [0, 1], lung-bbox + 20-voxel padding, kept at native voxel grid.
- **Training input** 128^3 patches sampled via `RandCropByPosNegLabeld` at a ratio of `pos:neg = 2:1`, `num_samples = 4` per volume.
- **Inference** MONAI `sliding_window_inference`, ROI (128, 128, 128), `overlap = 0.5`, `sw_batch_size = 4`, Gaussian blending, single output at the input resolution.
- **Output** (2, H, W, D) softmax logits: 0 = background, 1 = nodule.

2. Model

Property	Value
Architecture family	DynUNet / 3D U-Net
Spatial dims	3
Input channels	1
Output channels	2 (softmax)
6-level auto-configured	strides=[1,2,2,2,2], kernel=3
Trainable parameters	16,538,530

3. Data

Same unified corpus + split as the resize-recipe 256^3 models:

Source	Role
NLST	train + val + test
NSCLC-Radiomics	train + val + test

Source	Role
LIDC-IDRI	train + val + test

- **Split:** `unified.json` — patient-grouped, dataset-stratified. 1 683 train / 297 val / 325 test series.
- **Lung crop:** per-series 3D bbox from `bboxes_unified.json` (lung ROI + medium-model fallback), + 20-vox padding. No resampling.
- **Class imbalance:** nodule voxels are $\sim 10^{-5}$ of the total — see §7.

4. Training

Setting	Value
Loss	Focal Tversky + weighted CE ($\alpha=0.3$, $\beta=0.7$, $\gamma=2.0$, $\lambda_{CE}=0.1 / 0.3$, nodule class weight = 100)
Optimizer	Adam
Learning rate	1×10^{-5}
Weight decay	1×10^{-5}
LR schedule	Cosine annealing, $T_{max} = 400$, $\eta_{min} = 1 \times 10^{-6}$
Batch size	2 volumes $\times$ 4 patches = 8 patches / step
Patch size	128^3 (positive-biased random crop, <code>pos:neg = 2:1</code>)
Epochs	400
Random seed	42
Mixed precision	bf16 (autocast, no GradScaler)
Augmentation	3D flips, 90° rotations, elastic rotation, zoom, intensity scale/shift, Gaussian noise/blur, contrast
Hardware	1 $\times$ NVIDIA H100 80 GB (fat01)

Best checkpoint recorded at **epoch 264 / 400** (see `_meta` in metrics JSON).

5. Evaluation protocol

Evaluated on the `unified` **test** split (325 series held out — never seen during training or model selection). Predictions taken as `argmax` over the 2-channel softmax output from `sliding_window_inference`. All voxel-level metrics are micro-averaged over the whole split.

6. Results

Ticket-required metrics

Metric	Value
mean IoU	0.6557
Accuracy	0.9985
Precision	0.3379
Recall	0.8086

Supplementary

Metric	Value
IoU (foreground / nodule class)	0.3129
IoU (background class)	0.9985
Dice / F1 (micro)	0.4767

Per-case Dice distribution (325 test cases)

Statistic	Value
Mean	0.4219
Std	0.2577
Min	0.0000
p05	0.0047
p25	0.2078
Median	0.4386
p75	0.6311
p95	0.8233
Max	0.9219

Voxels evaluated: 7,849,317,259. Cases: 325.

7. Notes on interpretation

Sliding-window inference at 128^3 trades global context (a 256^3 resample sees the whole lung in one pass) for preserved resolution (no down-sampling of small nodules). On this unified corpus the trade tends to be neutral-to-negative on nodule Dice — small-nodule gains do not compensate for lost global context on typical test cases.

Class-imbalance caveats are unchanged from the 256³ reports: Accuracy is trivially ~1, mIoU is dominated by IoU_bg, and Dice / F1 (micro) is the honest voxel-level summary. Per-case mean weights each patient equally regardless of nodule volume.

Instance-level metrics (per-nodule)

Connected-component analysis with 26-connectivity, pooled over the held-out test split, on the native voxel grid.

- **Instance recall** — GT nodules touched by at least one predicted voxel, over all GT nodules.
- **Instance precision** — predicted components touching at least one GT voxel, over all predicted components.

Metric	Value
GT nodules (components)	947
GT nodules hit	803
Instance recall	0.8479
Predicted components	7,869
Predicted components hitting GT	827
Instance precision	0.1051

No minimum-size filtering is applied — every predicted component counts, so single-voxel false positives lower instance precision. Produced by `scripts/instance_metrics.py --config configs/dynunet_sw.yaml --ckpt <ckpt> --split test` (JSON: `results_instance/dynunet_sw.json`).

8. Reproducibility

- Config `/home/werner/nodule-training-demo/configs/dynunet_sw.yaml`
- Checkpoint `stage2_fine/checkpoints_dynunet_sw/best_model.pth`
- Metrics JSON `/tmp/claude-1002/-home-werner-nodule-segmentation/c2faa0a0-5a75-48f5-8288-377df40092b7/scratchpad/sw_metrics/dynunet_sw.json`
- Command `python stage2_fine/eval_metrics.py --config /home/werner/nodule-training-demo/configs/dynunet_sw.yaml --checkpoint stage2_fine/checkpoints_dynunet_sw/best_model.pth --split test`